

Cross-session price discovery in gold

**The Asian state, the London handover,
and the evidence that survived**

Asian market state -> London behaviour

A study of compression, directional displacement and the transfer of volatility in XAUUSD, with a controlled test of the Tokyo Coil idea.

Research conclusion: no validated profitable trading strategy was established. The empirical results reject compression-to-expansion as defined in the master prompt. Both fixed Coil transfer diagnostics fail validation.

Prepared for the human researcher and author of the XAUUSD Research project.

Research and implementation assistance: Codex.

4 October 2026 | Study version gold-coil-v1

Evidence status: exploratory analysis of a public archive claiming Dukascopy provenance. Original-provider authentication, an independent feed and the historical news calendar are unavailable. Studied period: 2016-2023. Reserved holdout: 2024 onward, unopened.

Historical replay, not live trading. The gold rules and results below are distinct from the USDJPY claims in TokyoCoil.pdf.

1. The result in one minute

Quiet Asia predicts quieter London. Breakouts alone do not prove an edge.

The primary study asks whether the Asian session contains useful information before London. It tests six registered hypotheses before constructing any new trading rules. Two independently specified Tokyo Coil diagnostics provide a transfer check, with explicit changes for gold and a missing news calendar.

Validation, 2022-2023	Fixed 08:00 NY exit	Noon NY diagnostic
Trades	265	265
Net expectancy, R/trade	-0.140	-0.094
Profit factor, R basis	0.74	0.87
Scenario compounded return	-27.75%	-29.18%
Closed-balance drawdown	30.82%	36.77%

A trade R is net hypothetical profit divided by initial price-stop risk. Returns assume a \$100,000 account, continuous-ounce sizing, 1% risk and the original causal equity brake. They are scenario results, not a broker-verified account simulation.

H1 shows a stable positive relationship between Asian and London volatility, opposite to the registered compression-to-expansion prediction. H2 confirms that smaller boxes are more frequently breached, but its nonmechanical break-distance test fails. Direction, reversal, interactions and regimes do not pass the evidence gate.

Decision: publish the failed hypotheses and research infrastructure. Preserve final holdout data. Do not present a profitable gold strategy on the strength of this study.

2. The logic: what should be true?

Tokyo Coil motivates a useful question: can a quiet regional session offer inexpensive risk before a later move? Its USDJPY performance claims are reference material, not evidence about gold. Gold is a different market; the transferred idea must be tested from first principles.

A small stop is a price, not an advantage

A narrow Asian box makes the opposite-edge stop cheaper. It also makes both boundaries easier to touch. The relevant economic question is whether favorable travel after entry compensates for false breaks, spread, commission and adverse fills. Without that compensation, inexpensive risk is simply frequent small losses.

Volatility is not direction

A forecast of a large London range does not identify which side to buy. A session can move far in both directions and stop both breakout orders. Conversely, Asia may predict London volatility while containing no directional edge. A volatility forecast and a profitable spot-gold strategy are different claims.

The core research chain

Asia: range, realized volatility, recent-range percentile, signed standardized return and close location.

London: boundary breach, first breach, return, range, excess distance and post-break excursions.

New York: reserved secondary context, never part of the initial entry signal.

The statistical result supports volatility persistence in this archive. That is an economically interpretable finding, but it is not the registered quiet-Asia expansion effect and does not justify a new directional strategy. A new persistence hypothesis would require a separately registered future test.

Confounds: changing gold price, clustered volatility, macro news, the gap between Tokyo close and London open, and the mechanical geometry of a narrow box.

3. Data provenance: what was obtained

Role	Actual evidence status
Primary requested feed	Dukascopy direct M1 bid/ask or ticks: not acquired. Direct legacy requests returned HTTP 429; alternative provider URL returned HTTP 403.
Exploratory laboratory	Public Market-Data-Lab CSV archive, commit-pinned, claiming Dukascopy M1 bid/ask origin.
Independent feed	Not acquired. A mirror of the same claimed provider is not independent replication.
News calendar	Not supplied with the PDFs and not acquired as a verified historical series.
GC futures	Official exchange data not acquired; no GC result is claimed.

Archive repository: kevingtlin/Market-Data-Lab. Pinned commit: 922f83a60cc574e7395fb27397077288055a1ef6. Its metadata advertises history through 20 August 2026; that is an archive coverage statement, not verified latest-provider coverage. Only 2016-2023 prices were downloaded for this study.

Each file contains UTC epoch-millisecond timestamps and open, high, low, close, with bid and ask in separate monthly files. Raw bytes remain unchanged. Source URL, file size and SHA-256 are stored in each split manifest. There is no tick count or volume field.

Checksums prove the files used are reproducible; they do not prove the uploader exported correctly or that the price path is authentic. This distinction limits all acceptance claims, even statistically strong findings.

Official access references: Dukascopy Historical Data Export and historical data access guide. The latter documents credentialed requester-pays access; none was used here.

4. The quality audit

Audit quantity	Development 2016-2021	Validation 2022-2023
Bid source rows	3,156,480	1,051,200
Ask source rows	3,156,480	1,051,200
Both-side flat rows removed	1,033,827	343,892
Invalid quote rows	0	0
Suspicious quote rows flagged	733	2
Duplicate bid timestamps	0	0
Nonmonotonic bid timestamps	0	0
Valid Asia sessions	1528	515
Valid London sessions	1546	515

Flat source candles during closures can make a calendar look perfectly complete. Both-side flat M1 rows are excluded as possible padding. With no native tick count, this deliberately conservative rule also removes genuine quiet minutes. Missing active minutes are audited rather than interpolated.

An analysis session needs at least 98% of expected active M1 minutes, observed first and final minute, no gap over five minutes, and no flagged quote. Nonpositive prices, invalid OHLC ordering or nonpositive endpoint spread are rejected. Spread over \$5/oz or adjacent midpoint-open jump over 2% is flagged; affected sessions are excluded.

Holiday names are not inferred from gaps. A closure or partial holiday session fails the same completeness rule. Daily audit CSVs record active minutes and gap sizes, including weekends. Excluding future-incomplete execution paths is an ex-post data-cleanliness choice; it can introduce selection bias and must be checked against a raw-tick replica.

Bid highs/lows define all range extrema. Midpoint open/close is the average of side endpoints. Averaging independent bid and ask highs/lows cannot reconstruct synchronized midpoint extrema; none is falsely asserted here.

5. Exact session clocks

Window	Local definition	UTC convention
Asia / Tokyo	09:00-15:00 Asia/Tokyo	00:00-06:00 year-round
London	08:00-12:00 Europe/London	08:00-12:00 GMT; 07:00-11:00 BST
New York	08:00-12:00 America/New_York	13:00-17:00 EST; 12:00-16:00 EDT

These windows were selected from the master prompt before predictive testing. They represent regional research periods, not official spot-gold exchange opening hours. IANA timezone conversion handles the applicable historical DST rules. Tokyo does not use DST.

The UK and US change clocks on different dates. Unit tests cover the March 2016 mismatch, winter/summer London, Tokyo, and the New York midnight-crossing order window. Fixed server offsets are not used in the Python engine.

Why Tokyo Coil is a separate benchmark

The reference actually forms its box at 20:00-23:00 New York and accepts a breakout at 23:00-03:00 New York. Those are Tokyo-hours entries, not entries triggered by London taking over. Its opening narrative and exact rules should not be conflated. The benchmark therefore cannot answer the primary Asia-to-London question by itself.

Primary valid-session coverage: development: Asia 97.6%, London 98.8% of 1,565 weekday dates; validation: Asia 99.0%, London 99.0% of 520 weekday dates. The first 20 valid sessions of each split supply feature warm-up and are absent from most regressions. Each split resets history, including the candidate brake; no continuity across split boundaries is implied.

Two alternative Tokyo research windows (00:00-05:00 and 01:00-06:00 UTC) are registered sensitivity checks on validation. They are not chosen by profit.

6. Experimental design and the holdout

Split	Dates	How used
Development	2016-01-01 to 2021-12-31	Registered H1-H6 tests and external benchmark evaluation.
Validation	2022-01-01 to 2023-12-31	Direction, magnitude, costs, regimes and registered robustness checks.
Final reserved holdout	2024-01-01 onward	Not downloaded or examined. No tradable effect passed the preceding gates.

The master prompt includes a phase-1-only starter. The user explicitly requested a full report, so the project proceeded through the controlled development and validation sequence while retaining the evidence gates. The instruction to manufacture no profitable backtest takes precedence over the desire for a profitable conclusion.

Seven unique primary statistics cover H1-H6; momentum and reversal share one coefficient with competing signs. The Benjamini-Hochberg procedure controls the false-discovery-rate family at 5%. Regressions use heteroskedasticity/autocorrelation-consistent standard errors with five lags, plus 2,000 circular five-session block-bootstrap draws.

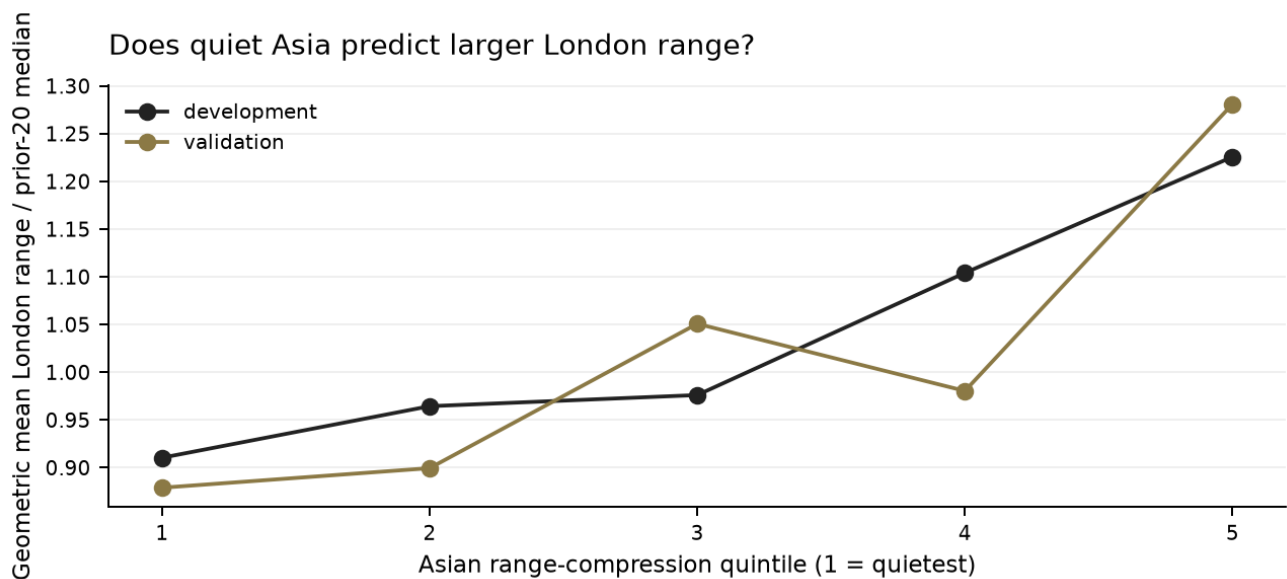
A promising effect needs development $q < 0.05$, expected sign, a predeclared economic floor, and validation $p < 0.05$ retaining at least half the development magnitude. Adequate samples are required. Original-feed authentication and execution validation remain additional gates.

The later user-requested percentile, RV and close-location tests are labeled exploratory because the first outcomes were already known. They cannot retroactively be called preregistered confirmation.

Frozen local code commit: e066e53c81d080dad09d6a764a32ff5ee283b837. Tests passed before recording the freeze. A guarded downloader and one-view marker support holdout discipline; they are logical controls, not an inaccessible external vault. The final holdout remains reserved.

7. Compression, expansion and breakouts

Test	Dev beta / q	Val beta / p	Interpretation
EXP001 N=1508/495	+0.288 / 2.3e-21	+0.322 / 6e-10 CI [+0.220, +0.424]	Opposite H1 sign
EXP002a N=1508/495	-0.238 / 1.4e-24	-0.192 / 7.9e-08 CI [-0.263, -0.122]	Narrow-box mechanics
EXP002b N=1508/495	-0.084 / 0.25	+0.156 / 0.15 CI [-0.057, +0.368]	No stable excess travel



Source: pinned archive, development 2016-2021 and validation 2022-2023. Quintile cutoffs are within-split visual diagnostics; regression tests remain continuous.

H1 regresses log normalized London percentage range on log Asian compression, controlling for lagged volatility and prior absolute Asian return. A negative slope would support compression-to-expansion. Validation beta is +0.322; its 95% block-bootstrap CI is [+0.216, +0.418]. The positive sign supports persistence instead. Table intervals are 95% HAC intervals.

H2 uses a breach-probability linear model and a separate excess-distance model normalized by the prior Asian-range median. The probability finding is strong but mechanically unsurprising. The distance coefficient is not significant and flips sign in validation. H2 therefore fails as a nonmechanical tradable effect.

8. Direction, reversal and interactions

Test	Dev beta / q	Val beta / p	Interpretation
EXP003_004 N=1508/495	-0.002 / 0.93	-0.054 / 0.33 CI [-0.161, +0.054]	No direction edge
EXP005 N=1508/495	-0.015 / 0.49	-0.007 / 0.95 CI [-0.233, +0.219]	No interaction evidence
EXP006a N=1448/435	+0.082 / 0.25	-0.092 / 0.37 CI [-0.295, +0.110]	No stable vol interaction
EXP006b N=1508/495	+0.100 / 0.24	+0.003 / 0.98 CI [-0.206, +0.212]	No stable trend interaction

D is the Asian log return divided by the standard deviation of the previous 20 valid Asian returns. The outcome is London log return divided by its own previous-20 standard deviation. Momentum requires a positive coefficient; reversal requires a negative coefficient of meaningful size. Neither competing explanation passes development and validation.

Among 68 validation sessions with $|D| \geq 1.5$, mean sign-adjusted London return is -0.139 outcome standard deviations; the 95% block-bootstrap interval is [-0.517, +0.202]. Continuation hit rate is 50.0%. That interval includes zero.

H5 tests the interaction between D and log compression with both main effects present. H6 tests volatility and trend interactions in the range model. The volatility regime uses lagged Asian range against its prior-60 baseline; the trend score uses only the previous 20 Asian returns. Missing warm-up regimes stay missing, not falsely low volatility.

HAC intervals protect against some serial dependence, not all regime shifts. A circular-shift permutation is an additional directional diagnostic, not an independent market replica. Rank correlation is useful for monotone effects and does not prove causality.

Decision: H3/H4/H5/H6 are inconclusive for tradable predictive information in this sample. Failure to reject zero is not proof that every possible effect is zero. It is a reason not to derive a trading rule from these tests.

9. The additional Asian-state diagnostics

Exploratory test	Development beta / q	Validation beta / q
E01_RV_compression	+0.466 / 1e-37	+0.564 / 1.1e-17
E02_close_location	-0.014 / 0.92	-0.055 / 0.84
E03_RV_direction	-0.028 / 0.33	-0.058 / 0.84

Prior-20 range state	N	Mean London norm. range	Boundary breach
Bottom 10%	72	0.93	97.2%
Bottom 20%	122	0.96	95.9%
Bottom 30%	170	0.97	96.5%

These tests were added at the user's request after viewing the original results. Range percentiles compare the current width against previous valid sessions only. The 10/20/30% states are nested, so they are not independent experiments. No most-profitable percentile is selected.

RV compression also predicts London range positively, consistent with volatility persistence. Close location, measured inside the Asian bid range, and RV-by-direction interaction do not establish a directional effect.

Validation first-boundary counts: up 232, down 225, ambiguous 0, none 58. M1 double touches remain ambiguous.

For 437 unambiguous paths with a lagged scale, median time to first breach is 7 minutes. Median post-break MFE is 0.51 and MAE 0.31, each in prior-median Asian-range units.

Excursions are bid-extrema path diagnostics measured from the boundary, including the first bar; they are not attainable profits or tick-verified trading returns. M1 cannot reveal whether its adverse or favorable extreme occurred first.

10. The exact rules: diagnostic benchmarks

No new trading strategy passed the research gate.

The following rules evaluate the externally supplied Coil concept. They are fully specified so another researcher can reproduce a failed transfer test. They are not a recommendation or an accepted gold strategy.

- 1. Box.** Sunday-Thursday New York evening, bid high/low over 20:00-23:00; at least 30 M5 bins with four active M1 observations each; width at least \$0.20/oz. The gold floor is an engineering assumption, not the reference's two USDJPY pips.
- 2. Coil.** Width below the median of valid boxes among the previous 20 eligible weekday sessions, with at least ten prior valid boxes. This follows the reference code's session-based lookback; the prose's "20 valid boxes" differs slightly.
- 3. Entry.** First bid-edge breach between 23:00 and 03:00 NY. Long uses threshold/open plus observed minute-open spread and adverse slippage; short uses threshold/open minus slippage. One trade; cancel the other side. Dual-edge first-touch minute skips the day.
- 4. Stop.** Opposite bid edge for long; opposite high plus entry-minute observed spread for short. No target or trailing stop. Same-minute stop touches and adverse gaps count against the trade.
- 5. Exit.** Two predeclared variants: fixed 08:00 NY, or fixed noon NY. Neither reproduces Tokyo Coil's historical news-calendar flattening. Missing path or exit quotes invalidate the replay day.
- 6. Risk.** 1% balance to initial stop; half while prior 20 closed R sum below -4. Continuous ounces; lot/margin constraints unverified. One position, no intentional 17:00 NY rollover.
- 7. Costs.** Bid/ask endpoints, assumed \$0.07/oz round-trip commission and \$0.05/oz adverse slippage per fill. Intraminute synchronized spread is unknown. Entry spread approximation is explicit, not claimed exact execution.

11. Worked examples from actual archive rows

Example / NY evening	Direction / entry UTC	Exit UTC / reason	Net R
Representative winner 2023-10-09	short 2023-10-10 04:29:00+00:00	2023-10-10 12:00:00+00:00 time	0.856
Representative loser 2022-11-06	short 2022-11-07 06:33:00+00:00	2022-11-07 08:54:00+00:00 stop	-1.022
Worst replay outcome 2023-08-29	short 2023-08-30 05:52:00+00:00	2023-08-30 09:21:00+00:00 stop	-1.047



Source: pinned archive, 2023-10-09 NY evening; 08:00-exit diagnostic. The line is quoted minute-open midpoint, not the full intramminute path.

These are hypothetical replay trades, never broker executions. Winner and loser are selected nearest the median result within their sign groups; the unusual example is the lowest net R. This selection rule avoids showcasing a spectacular winner as a normal outcome.

The winner's box spans 1861.295-1865.315 USD/oz; relative width 0.925. Initial price-stop risk is 4.360 USD/oz. Its +0.856R is net of the stated cost scenario. A stop can lose more than 1R when modeled slippage or a gap worsens the fill.

12. Development, validation and unseen status

Metric	Dev 08:00	Dev noon	Val 08:00	Val noon
Trades	732.00	734.00	265.00	265.00
Total net R	-28.62	10.23	-37.07	-24.99
Profit factor (R)	0.92	1.02	0.74	0.87
Return (%)	-24.69	3.71	-27.75	-29.18
CAGR (%)	-4.62	0.61	-15.03	-15.88
Daily Sharpe	-0.34	0.13	-1.50	-0.92
Daily Sortino	-0.56	0.28	-2.24	-1.84
Win rate (%)	41.12	32.43	37.36	26.04
Expectancy (R/trade)	-0.04	0.01	-0.14	-0.09
Avg winner R	1.11	2.08	1.08	2.45
Avg loser R	-0.84	-0.98	-0.87	-0.99
Closed DD (%)	34.20	34.39	30.82	36.77
Cash-PnL profit factor	0.91	1.01	0.72	0.80

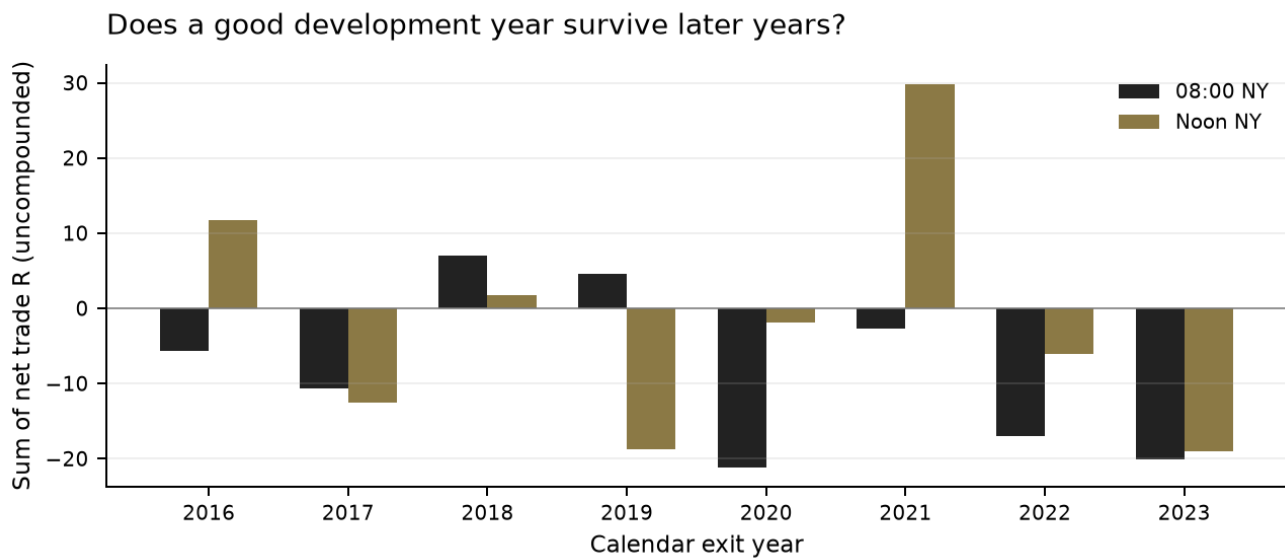
Development: 2016-2021. Validation: 2022-2023. Both fixed exit variants were declared before their results. The noon diagnostic is approximately flat in development and loses in validation; the early-exit adaptation loses in both. Neither supports a validated profitable-strategy claim.

Profit factor is the sum of positive R divided by the magnitude of negative R. Sharpe uses daily closed returns, including zero-return business days, with $\sqrt{252}$ annualization and a zero risk-free-rate convention. Sortino uses root mean squared negative daily returns, likewise including zero days.

The final unseen 2024-onward holdout is **not run**. That is a preserved test, not a missing positive result. Negative validation and failed mechanism gates make consuming it unjustified for these candidates.

The report does not infer trading returns for October 2026 or any other untested interval. Return and drawdown here refer to the sizing scenario and closed balance; intraday mark-to-market drawdown, financing tariffs and executable broker constraints have not been measured.

13. The year-by-year result

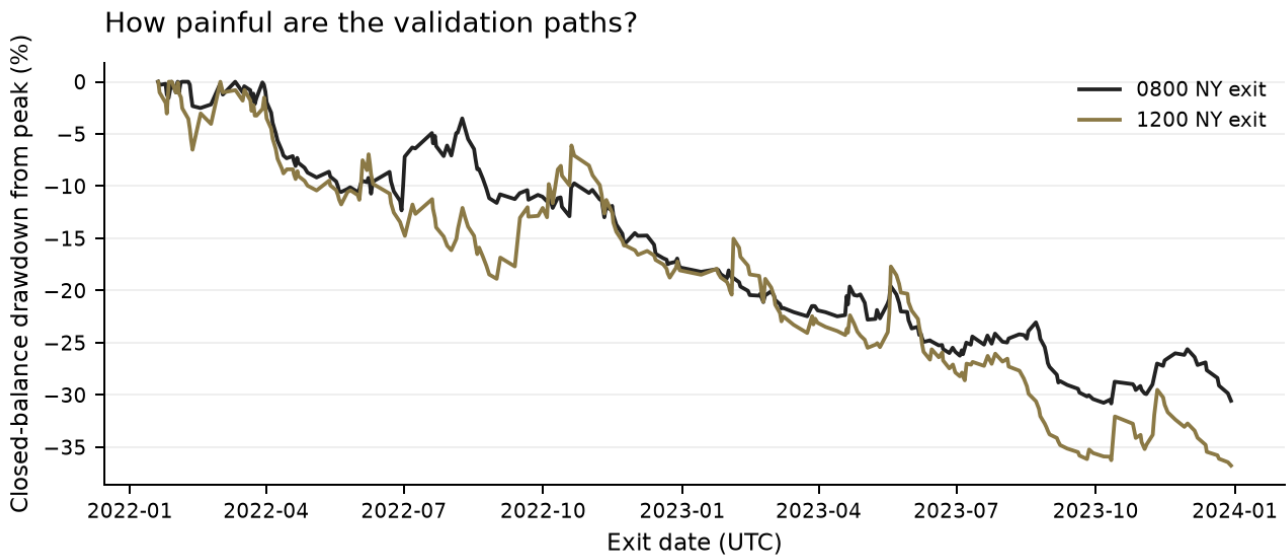


Source: pinned archive, 2016-2023. Risk brake affects compounded money returns, not the sum of trade R.

Exit year / split	08:00 N / net R	Noon N / net R
2016 / development	118 / -5.69	118 / +11.71
2017 / development	126 / -10.62	126 / -12.56
2018 / development	119 / +7.05	119 / +1.78
2019 / development	123 / +4.55	125 / -18.71
2020 / development	120 / -21.19	120 / -1.86
2021 / development	126 / -2.71	126 / +29.87
2022 / validation	136 / -16.95	136 / -6.02
2023 / validation	129 / -20.12	129 / -18.96

Profitable years occur inside an unprofitable validation process. They are not evidence that a trader can identify those years in advance. No year, date or regime is deleted because it worsens the story.

14. The pain: losses, flat periods and tails



Source: validation 2022-2023, \$100,000 initial, 1% base risk and causal brake. Open-position drawdown is absent, so this understates full execution risk.

The early-exit diagnostic's longest validation losing streak is 10 trades; the noon diagnostic's is 19. Their maximum closed-balance drawdowns are 30.8% and 36.8%. Longest daily closed-balance underwater spells are 468 and 477 business days.

Positive-month frequency is 45.8% versus 33.3%. Both validation years are negative under each diagnostic. A few favorable trades coexist with sustained negative expectancy; waiting longer is not demonstrated to solve it.

Other validation metrics, early/noon respectively: annualized daily volatility 10.17% / 16.66%; exposure 1545.4 / 1935.0 hours, or 12.38% / 15.50% of a 24-hour business-day calendar; two-sided turnover 75,718 / 81,914 ounces. Payoff ratios are 1.25 / 2.47.

Holding longer raises the size of the occasional winner, but also raises variance and losing-streak risk. The equity brake uses past losses to reduce size; it can limit some damage but cannot turn a negative underlying expectation into a validated edge.

Spread spikes, macro events, unavailable ticks, venue dependence and sudden gaps remain material tail risks. They are not erased by a one-trade-per-day rule or a no-rollover exit.

15. Robustness: neighboring choices

Threshold / shift	08:00 PF / net R	Noon PF / net R
threshold 0.8	0.90 / -8.1	1.25 / +25.0
threshold 0.9	0.79 / -23.5	0.95 / -7.9
threshold 1.0	0.74 / -37.1	0.87 / -25.0
threshold 1.1	0.82 / -27.8	0.94 / -12.7
threshold 1.2	0.84 / -25.9	0.96 / -9.8
box_shift -30.0	0.77 / -33.3	0.98 / -2.9
box_shift 30.0	0.77 / -32.2	0.99 / -1.6

Only validation is used for registered parameter sensitivities. Five compression thresholds and two box shifts were tried; none was selected as a new strategy. Shifted box ends and entry starts move together, preventing an entry from using a still-unfinished box.

The favorable cell is disclosed: threshold 0.8 with noon exit earns +25.04R on 140 validation replay trades, profit factor 1.25 and mean +0.179R. Its exploratory 10,000-draw five-trade block-bootstrap 95% expectancy interval is [-0.182, +0.561]R. It includes zero. Selecting this cell after seeing validation is not evidence of a confirmed profitable strategy; it requires a new clean selection/evaluation protocol.

Validation early-exit long trades total -18.95R; shorts total -18.12R. Direction, weekday and regime slices are descriptive diagnostics, not permission to select a profitable subgroup.

Alternative research-window outputs are saved separately. These checks preserve the original hypothesis meanings, rather than rewriting quiet-Asia expansion into any statistically convenient relationship.

The package removes the top 1%, 5% and 10% best trades and records expectancy intervals and return serial dependence. These diagnostics remain in the bundle.

A parameter surface is useful when neighboring rules gradually change the same phenomenon. It cannot rescue failed validation by picking a single favorable cell after seeing its return.

16. Costs and slippage sensitivity

Cost / slippage scenario	08:00 PF / E[R]	Noon PF / E[R]
cost 1.5	0.67 / -0.184	0.84 / -0.120
cost 2.0	0.61 / -0.229	0.79 / -0.160
cost 3.0	0.53 / -0.281	0.73 / -0.208
slippage 0.0	0.78 / -0.120	0.90 / -0.074
slippage 0.1	0.71 / -0.159	0.85 / -0.114
slippage 0.25	0.63 / -0.214	0.77 / -0.172

Baseline includes quoted side prices, \$0.07/oz round-trip assumed commission and \$0.05/oz adverse slippage each fill. Cost stresses scale modeled spread and commission to 1.5x, 2x and 3x; slippage is stressed separately at \$0, \$0.10 and \$0.25/oz per fill. The baseline \$0.05 is already shown in the main results.

Empirical validation spread diagnostics: asia: median session-median spread \$0.347/oz; 95th percentile \$0.444/oz; london: median session-median spread \$0.332/oz; 95th percentile \$0.404/oz; new_york: median session-median spread \$0.330/oz; 95th percentile \$0.407/oz. Each statistic describes observed endpoint quotes in qualifying sessions, not a universal gold spread.

Widened spread stress is a conservative modeling scenario, not a reconstruction of synchronized bid/ask quotes. Gaps are filled adversely and commission is debited even after a stop. Broker minimum lots, contract size, leverage and execution rejection must be added for a broker-accurate account test.

A system that fails before these additional execution frictions should not be sold as profitable because a chart-only or zero-cost backtest looks better.

17. Monte Carlo: what the experiment means

Monte Carlo samples many possible future paths from the observed validation return distribution. It estimates the consequences of that empirical distribution under the specified sizing rule; it cannot invent an edge, prove independence or predict events absent from the sample.

Two models, both explicitly simulated

Trade bootstrap: sample validation trade R independently with replacement, using the empirical annual trade count across 252 business days. This discards serial dependence and approximates timing by equally spaced trade slots.

Block bootstrap: sample circular blocks of ten consecutive observed validation business-day returns, including zero-trade days. This preserves some clustering and event spacing within each block.

Every path recomputes the causal equity brake from its own last 20 trades. Four base risks are tested: 0.25%, 0.50%, 0.75% and 1.00%. There are 10,000 paths per method, risk and exit candidate: 160,000 one-year paths total. Initial equity is \$100,000, seed 20261004, horizon 252 business days.

The reported probabilities are simulation frequencies, not objective probabilities of next year. At 10,000 draws, sampling error is largest near a 50% frequency (roughly half a percentage point standard error). A zero observed ruin frequency does not mean zero real-world ruin risk.

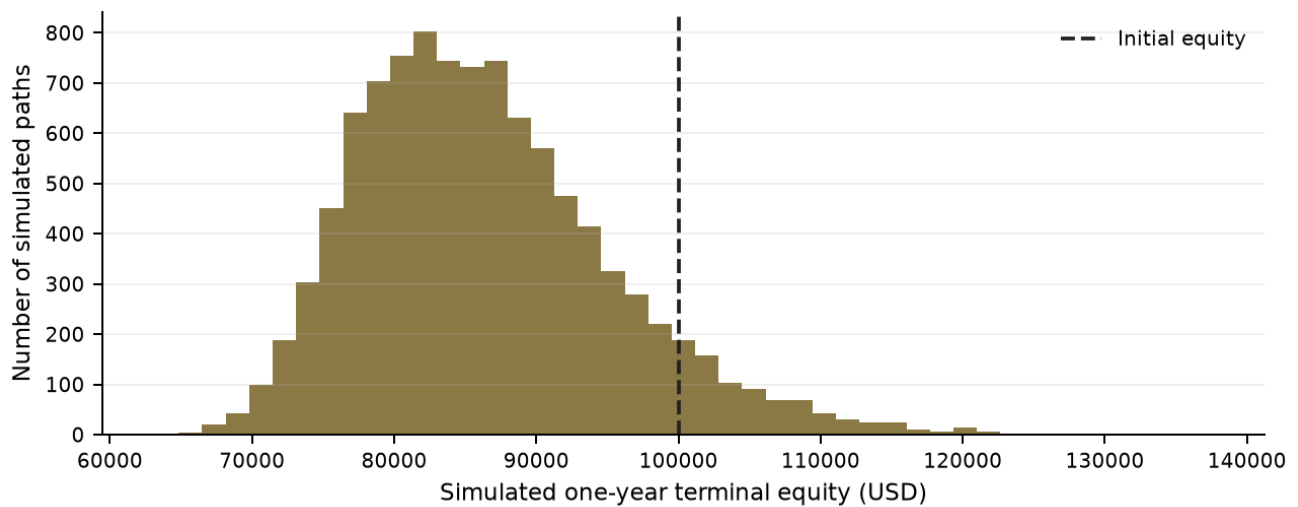
Monthly diagnostics use twelve 21-business-day buckets, not real calendar months. Ruin is defined as equity falling to 50% of the initial account, distinct from broker liquidation. Closed-balance drawdown omits open-position adverse excursions.

The package saves terminal-equity, annual-return, Sharpe, drawdown, drawdown-duration, losing-streak and losing-bucket distributions. Block risk is the main display; iid trade results remain available for comparison.

18. Monte Carlo: the risk results

Base risk	P(loss year)	P(DD>20%)	DD 95th pct	Equity 5th pct
0.25%	91.4%	0.00%	7.7%	\$92,804
0.50%	91.6%	0.01%	14.8%	\$86,081
0.75%	91.9%	10.41%	21.4%	\$79,808
1.00%	92.2%	43.71%	27.6%	\$73,953

What does the failed early-exit distribution imply?



Source: validation 2022-2023 empirical R; 10,000 simulated ten-business-day block paths; 1% base risk plus brake. This is simulated risk, not historical profit.

At 1% base risk, the block model gives $P(DD > 10\%) = 96.1\%$, $P(DD > 20\%) = 43.7\%$ and $P(DD > 30\%) = 1.4\%$. Expected longest losing streak is 8.7 trades. Observed simulated 50%-initial-equity ruin frequency is 0.00%.

Reducing risk reduces monetary loss and drawdown, not the sign of the underlying expectancy. No risk level is selected for live use. The noon candidate has its own parallel simulation files; using its larger winners to ignore its wider tails would be misleading.

19. Independent replication and execution

Claim	Status / required evidence
Second XAUUSD feed	Not performed. Need an independently operated broker feed and verified timestamps, side quotes and history completeness.
MT5 real-tick replay	Not performed. Need trade-by-trade matching of date, direction, entry time, costs and exit; identify generated-tick fallback.
COMEX GC economic cross-check	Not performed. Need official intraday history and documented contract/roll handling; compare effect signs, not identical spot prices.
Historical event calendar	Not available. Exact original Coil news flattening cannot be reproduced.

A second file downloader or mirror is not an independent venue. The source limitation applies to statistical effects and trade replay alike. Repeating the same archive with another program is a software correctness check only.

MetaTrader documentation notes that real-tick testing can generate ticks for intervals where real ticks are unavailable. A tester label therefore does not, by itself, establish complete real-tick coverage. Record the gaps and the actual replay mode per interval.

Official references: MetaTrader testing on real ticks; OANDA candle definitions; CME DataMine. These are data-access and methodology references, not sources of results in this report.

No EA was deployed and no live trade was placed. Writing an MT5 executable before resolving the failed research and data gates would imply readiness the evidence does not support. The delivered implementation is a reproducible Python research runner.

20. Failed research and rejected narratives

Hypothesis / narrative	Decision	Reason
H1 quiet Asia -> expansion	REJECT, registered direction	Positive volatility slope in both splits, opposite to the required negative sign.
H2 quiet Asia -> tradable break	INCONCLUSIVE / gate fail	Probability passes; normalized travel fails and changes sign.
H3 momentum / H4 reversal	INCONCLUSIVE	Shared directional test cannot establish either explanation.
H5 compression x direction	INCONCLUSIVE	Interaction lacks significance and stable magnitude.
H6 regimes rescue effect	INCONCLUSIVE	Registered interactions fail; no regime mining permitted.
Tokyo Coil transfer	REJECT as validated strategy	Both fixed diagnostics fail net validation; calendar and independent replication absent.

The development noon benchmark earns a small positive sum of R, but loses in validation. This is exactly why a profitable in-sample equity curve is not enough. Neither changing the exit after seeing validation nor shifting clocks toward a favorable sample would count as a clean discovery.

The positive persistence effect is retained as a research observation. Calling it a successful H1 after registering the opposite sign would be post-hoc relabeling. It can motivate a new independently registered study, with fresh future evaluation, but cannot certify the failed trading benchmark.

Failed experiments and their outputs remain in the bundle. No synthetic prices, invented event dates, fabricated live executions or nonexistent historical sessions are used to fill the gaps.

21. Implementation appendix and portfolio context

Equations and interpretation

Return: $r = \ln(P_{\text{close}} / P_{\text{open}})$. Log returns add across sequential periods; they are dimensionless.

Range: Asian bid high minus bid low, in USD/oz. Percentage range divides by midpoint open.

Realized volatility: square root of the sum of squared M5 midpoint close-to-close log returns; the first observation uses the initial complete M5 open. It measures observed variation, not implied volatility.

Compression: current Asian dollar range / median prior 20 valid Asian dollar ranges. A value below 1 is smaller than that trailing median.

Displacement: Asian log return / standard deviation of prior 20 valid Asian returns.

Close location: (Asian bid close - Asian bid low) / Asian bid range.

Probability, estimation and uncertainty

Expectation is a probability-weighted average; sample mean R estimates it. Variance measures dispersion and covariance measures co-movement. A confidence interval quantifies sampling uncertainty under a model, not a guarantee of future return. A p-value measures how incompatible the statistic is with its null, not the probability the null is true. Multiple testing raises false-positive risk; BH q-values account for the declared family.

Regression estimates conditional association, not causation. HAC errors allow specified residual dependence; block bootstrap preserves short contiguous clusters. Stationarity would mean a stable distribution over time; changed gold regimes can violate that approximation. Walk-forward validation respects time order but cannot eliminate source bias or post-hoc selection.

Registry, files and portfolio limits

EXP001 compression/range; EXP002a/b breach probability/excess distance; EXP003_004 competing direction; EXP005 interaction; EXP006a/b regimes; EXP007 external benchmark; EXP008 sensitivities; EXP009 Monte Carlo. Full primary regression mean, median, standard deviation, skewness, excess kurtosis, correlations, HAC t/p/CI and bootstrap CI are saved in hypothesis CSVs.

Source PDFs: TokyoCoil.pdf (USDJPY reference; no independently verified performance claim) and XAUUSD_Quant_Research_Master_Prompt.pdf (research specification). Official source links appear on pages 4 and 20. There are no other sleeve returns, so no portfolio correlation or diversification benefit is claimed. No trading sleeve passes acceptance.

22. The research runner: setup and architecture

Component	Purpose
download.py	Commit-pinned monthly archive retrieval, raw-byte SHA-256 manifests and guarded holdout acquisition.
study.py	Quote audit, active-M1 cleaning, causal sessions/features, HAC and block-bootstrap tests, conservative replay and sizing.
addendum.py	Explicitly exploratory percentiles, RV state, close location and first-break London paths.
robustness.py	Registered validation sensitivities and empirical Monte Carlo with stateful brake.
freeze.py / tests	Passing timezone, path-order, cost, gap and brake checks; Git freeze record before any holdout access.
report.py	Generate this PDF and source-labeled figures from saved results.

Install the pinned requirements with Python 3.13. Keep raw inputs immutable and processed outputs separate. The bundle includes configurations, registries, environment versions and audit/trade ledgers; it excludes raw source CSVs from redistribution.

Engineering tests use clearly identified fabricated fixtures only to check behavior. Those fixture quotes are never included in research rows, charts, trade ledgers or Monte Carlo samples. Passing a unit test is not proof that the feed or trading hypothesis is valid.

Eight checks passed: local-session DST, US/UK mismatch, midnight crossing, ambiguous dual-edge entry, long ask/bid costs, adverse stop gap, causal equity brake and deterministic bootstrap.

C++ is deferred. It would be useful only after statistical and execution correctness, if profiling identifies a real replay or simulation bottleneck. No unmeasured speed-up is claimed.

23. How to reproduce and audit it yourself

- 1. Check the evidence status.** Read README.md, study.json and the experiment preregistration. Confirm the source archive commit and understand that this is exploratory, unauthenticated venue evidence.
- 2. Acquire development only.** Run the documented downloader for 2016-2021. Verify raw SHA-256 values against manifest_development.json. Preserve input bytes and inspect the daily audit before testing any outcome.
- 3. Rebuild features and tests.** Run the development study, then validation acquisition and study. Compare saved hypothesis coefficients, sample sizes and trade-level R to the report. Run addendum.py for the clearly labeled later exploratory diagnostics.
- 4. Challenge execution.** Pick the worked examples, reconstruct the NY box from source bid bars and compare entry, stop and exit timestamps. Confirm commission, side quotes and slippage. Do not infer the order of intraminute extremes from M1 bars.
- 5. Reproduce sensitivities and risk.** Run robustness.py using only validation. Confirm seed, 10,000 paths per scenario, ten-business-day blocks, 252-day horizon and simulated rather than historical output labels.
- 6. Preserve the reserved sample.** Do not run the holdout downloader merely to see if failed rules recover. If a new hypothesis is developed after learning these results, define a clean future or independently locked test. Reproduction of viewed history is not another unseen experiment.
- 7. Recreate the report.** Run report.py after saved outputs and freeze metadata exist. Requirements and exact commands are in README.md. All quantitative tables are generated from analysis files, not manually invented numbers.

To upgrade this research to primary evidence, replace the third-party archive through a separately versioned authenticated acquisition, verify the calendar and source semantics, rerun the audit, and document exactly which historical outcomes were already known.

24. Conclusion, confidence and future research

A defensible null result is more useful than a manufactured profitable strategy.

The studied archive supports Asian-to-London volatility persistence: sessions with more Asian volatility tend to have more London volatility. It does not support the master prompt's quiet-Asia expansion prediction, a robust directional continuation/reversal rule, or an economically validated compression breakout.

Both fixed Tokyo Coil transfer diagnostics lose in validation under the frozen cost scenarios. The honest answer to "then find a profitable strategy" is that this study has **not established one**. That is a result about these hypotheses and tests, not proof that no gold strategy can exist.

Confidence: strong within-archive evidence for the sign of volatility persistence; weak evidence for directional predictability; no demonstrated trade-ready edge. Venue authenticity, historical event coverage and tick execution limit external validity. The reserved 2024-onward sample is unexamined.

Next research that the evidence permits

Authenticate the source and repeat the mechanism tests on an independent XAUUSD feed. Register volatility persistence as a new question before translating it into any entry logic. Study incremental forecasts relative to an explicit lagged-volatility baseline, and evaluate net execution only after the statistical gate passes. Use fresh future data for any rules changed after this study.

Research disclaimer

This report describes exploratory statistics and hypothetical historical replay. It is not financial advice or a recommendation to trade. Past or simulated performance does not guarantee future results. Live spread, slippage, gaps, financing, margin and broker constraints can worsen results. No live account profitability, original Tokyo Coil replication or GC validation is represented.

The report preserves the user's central research idea and the reference's clear rules/results/pain format, while documenting every deviation and every uncompleted evidence gate.